

WASP-36b

R-1922

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_160202 · created 2026-08-06T00:54:23+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457420.8292 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.164 +/- 0.048
Transit depth (Rp/Rs)^2	2.69 +/- 1.57 [%]
Orbital Inclination (inc)	88.11 +/- 2.89
Airmass coefficient 1 (a1)	8.35 +/- 4.14
Airmass coefficient 2 (a2)	-1.53 +/- 1.09
Scatter in the residuals of the lightcurve fit is	53.0 %
Best Comparison Star	None
Optimal Aperture	2.29
Optimal Annulus	3.25
Transit Duration (day)	0.093 +/- 0.0083

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.164 ± 0.048 vs 0.1368 (deviation 0.57σ)
Duration fit vs published	0.093 d vs 0.0758 d (deviation 2.07σ)
Mid-time O–C	4.5 min
Depth SNR	3.4
Residual scatter	53.0 %

Light curve

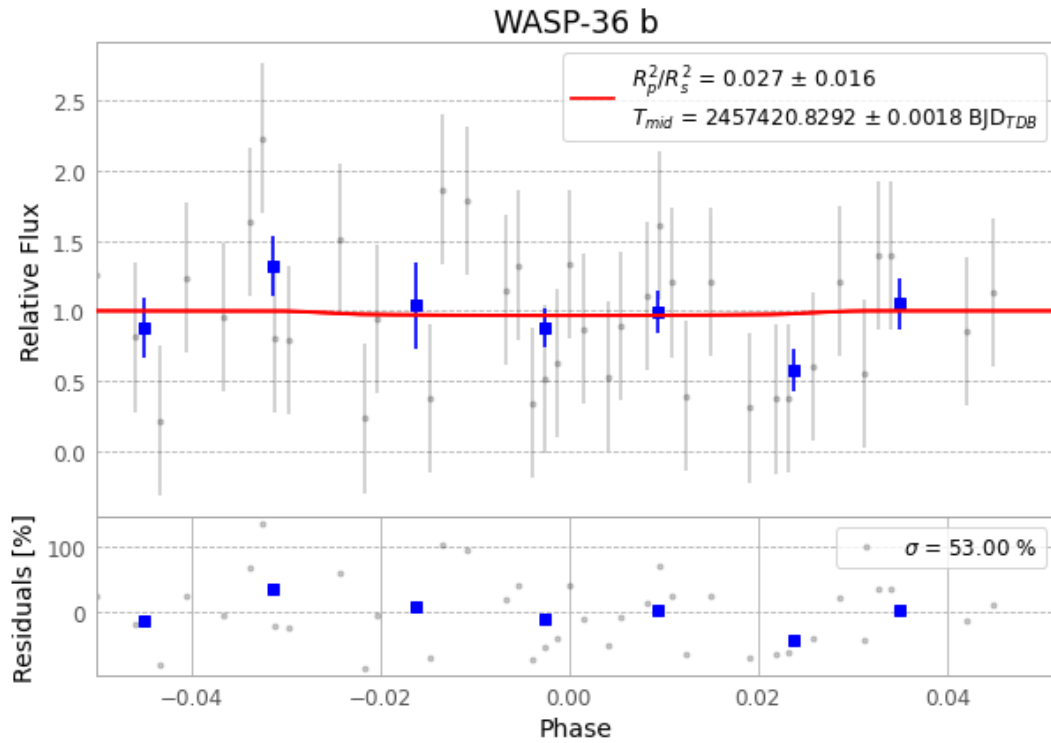


Figure 1 — FinalLightCurve_WASP-36 b_2016-02-01.png (SHA-256 a0e6b804c1a78b5f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [461, 42], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 550ae1f0f6c84b37...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-36160202055412.FITS → WASP-36160202094212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-160202.log (ba6901eb27f8...), FinalLightCurve_WASP-36 b_2016-02-01.png (a0e6b804c1a7...), FinalLightCurve_WASP-36 b_2016-02-01.csv (691e94c533cf...)

Compute resources

Wall time 187.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 00:54Z.